Double Reduction Gearbox Design Project

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1 Solution Table

1.1 Designer Chosen Parameters

Table 1: Shaft Specifications

Parameter	**Shaft a**	**Shaft b**	**Shaft c**
D (mm)	n/a	40	40
LL (in)	n/a	2.93	2.93
LR (in)	2.413	2.413	n/a
Material	n/a	303A	303A
Surface Type	n/a	machined	machined
Reliability	95.00%	95.00%	95.00%

Table 2: Gear Specifications

Parameter	**Gear B**	**Gear D**	**Gear C**	**Gear E**
Teeth	60	140	72	150
FOS	1	1	1	1
Reliability	95.00%	95.00%	95.00%	95.00%

Table 3: Bearing Specifications

Parameter	**Bearing C**	**Bearing D**	**Bearing E**	**Bearing F**
FOS	1	1	1	1

1.2 Assumptions

The following list names all assumptions made during the analysis and design of this double reduction gearbox that were not implicit assumptions of the course material and project. These assumptions were either standard machine design assumptions used to simplify calculations or designer decisions which were made to out of practicality.

- Conservation of energy, the system's efficiency is 100%
- Meshing angles do not split radial and tangential forces, instead they are felt directly by the shaft
- Transverse Shear $\ll$ Bending Stress so it was ignored during internal stress calculation, this avoids costly integration calculations which would slow the algorithm's speed
- There were ideal simple supports at bearings
- 95% reliability across all system objects, keeping failure probability uniform across all components

- The system was optimized for a design factor of 1.0, assuming that standard industry padding factors sufficiently account for environmental and material uncertainties
- The coordinate system used for the project was: X axis: left to right of shafts, Y: bottom to top, Z: in page to out, from system diagram

2 Introduction

Because of the design freedom of this project, an approach was implemented which allowed for total optimization of the system under the listed assumptions and for a specific parameter. Weight (or mass) minimization was chosen as the optimization variable as it was easy to implement and could be bridged to cost as generally weight and cost are positively correlated.

2.1 Methodology

A model Gearbox object was designed and implemented in python containing three objects: shaft, gear, and bearing. From there, A Genetic Algorithm was used which generated varying "DNA" sequences [N_b, N_c, N_d, N_e, LL, LR, D_b, D_c] and expressed them in a population of 100 Gearbox objects.

The result was scored by weight of the resulting system with the upper 50th percentile passing on their DNA (with a mutation rate of 3%). If a project constraint was violated, a mass of 999999 kg was assigned, eliminating it from the gene pool for the next generation.

Three conservative, "guaranteed-to-pass" bloodlines were seeded in the first generation as algorithmic boosts to accelerate convergence. Additionally, elitism strictly conversed the best result of every generation into the next to prevent backsliding and preserve the best DNA. Adaptive mutation rates also ensured local minimum were escaped by spiking mutation rates to 25% after 25 generations yielded no change in weight (decreasing again after a weight change).

2.2 Results

The algorithm was run with two 1000 total generations, taking 7-10 minutes to complete, each converging on the final design in the ∼150th generation (∼4 minutes of runtime as the spiked mutation rate lead to more designs failing earlier in the constraint validation phase, taking less compute time).

Either the optimizer reached a local minimum and failed to mutate out of it, or this result is the global minimum under my assumptions with the discreet component catalogues provided in the lecture slides. The resulting expressed Gearbox is my selected final design weighing 423.85 N (43.21 kg).

Many calculations of the optimizer were implemented to account for the general cases of a gearbox for N-number of components in arbitrary configuration (although other functions were hard coded specifically for this gearbox layout for the sake of scope creep). Therefore for the purpose of this project, a separate script - which used conservative, simple hard coded derivations - was made to test the output of the optimizer program.

The percent difference comparison between the conservative script and the optimizer is provided in the numerical results section.

3 Derivation and Sequence

Note: derivations follow the validation script's method, which follows the courses standard assumptions of no component (bearing or shaft) mass. The optimizer accounts for these factors as they are integral for optimizing weight. The percent difference in the numerical results section is evidence for why engineering courses neglect these factors as their effect is $\ll$ than the effects of gear interactions.

3.1 Gear Train and Gear Width

The first stage of validation involves ensuring the inputs and outputs of the system meet specifications through gear train math. The input rpm for me is 1050 rpm on shaft a and the output required is 216 rpm on shaft c. The pitch line velocity can also be derived from the angular velocity for use in finding factors.

$$\omega_c = \omega_a \left(\frac{N_B}{N_D}\right)\left(\frac{N_C}{N_E}\right) \quad ; \quad V = \frac{\pi d \omega}{12} \tag{1}$$

The stresses felt by the gear teeth can be equated to strength to find the required width to support the stress with a design factor of 1. Non-applicable strength factors are ignored in derivation (k_t).

$$\sigma = \frac{W_t P}{bJ} K_v K_o K_m \tag{2}$$

$$S_n = S_n' \cdot C_L \cdot C_G \cdot C_S \cdot k_r \cdot k_{ms} = (0.5 \cdot S_u) \cdot C_L \cdot C_G \cdot C_S \cdot k_r \cdot k_{ms} \tag{3}$$

Forces and moments at gears can be found from gear train math. The forces felt by the shaft which the gear is mounted on in $\mathbb{R}^3$ cartesian coordinates is found by reflecting the force applied by the gear to the meshing gear. The moments are found by using the cross product of the diameter (with directionality) with the forces to find the $\mathbb{R}^3$ moment vector, whose x component (according to the assigned coordinates in the assumptions section) is the torque.

$$T_b = T_a \left(\frac{N_D}{N_B}\right) \quad , \quad T_c = T_b \left(\frac{N_E}{N_C}\right) \tag{4}$$

$$W_t = \frac{2T}{d} \quad , \quad W_r = W_t \tan(\phi) \tag{5}$$

The above equations are the standard machine design equations derived to validate the vector math done in the program.

3.2 Reaction Forces, Shaft Stress, and Deflection

From the outputs of the program, a FBD can be constructed for shafts b and c.

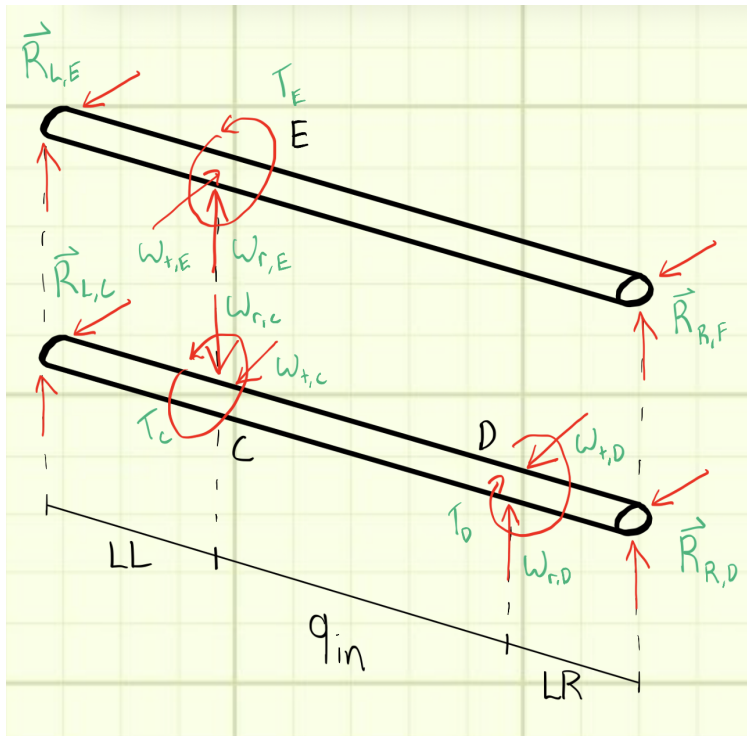

Figure 1: Drawn FBD of the forces at play on shaft b and c

3.2.1 Forces and Moments Equilibrium for Reaction Forces

Once forces and moments at the gears are known, Reaction forces at bearings can be found with static equilibrium equations. Example derivation is done on shaft b on the plane normal to the z-axis (the vertical components and moments in the y-x plane). It is important to translate the input LR or LL into a distance from whatever bearing you taking the moment about. Additionally, there are two planes to do the balances about, as the gears have two dimensions of effect.

$$\sum M_R = 0 = R_{Ly}(L) + W_{r,D}(L - LR) - W_{r,C}(LL) \tag{6}$$

$$R_{Ly} = \frac{W_{r,C}(LL) - W_{r,D}(L - LR)}{L} \tag{7}$$

$$\sum F_z = 0 = R_{Ly} + R_{Ry} - W_{r,C} + W_{r,D} \tag{8}$$

$$R_{Ry} = W_{r,C} - R_{Ly} - W_{r,D} \tag{9}$$

3.2.2 Internal Forces and Stresses

The shear, torsion, moment piecewise equations can be found by starting at one end and accounting for the forces felt by the shaft from each mounted component. Staying with shaft b for example and using singularity notation for the equations.

$$T(x) = T_C\langle x - LL\rangle^0 - T_D\langle x - (L - LR)\rangle^0 \tag{10}$$

$$V(x) = -R_{Ly}\langle x - 0\rangle^0 + W_{r,C}\langle x - LL\rangle^0 - W_{r,D}\langle x - (L - LR)\rangle^0 \tag{11}$$

$$M(x) = -R_{Ly}\langle x - 0\rangle^1 + W_{r,C}\langle x - LL\rangle^1 - W_{r,D}\langle x - (L - LR)\rangle^1 \tag{12}$$

Here are the resulting graphs for shear, moment, and torsion.

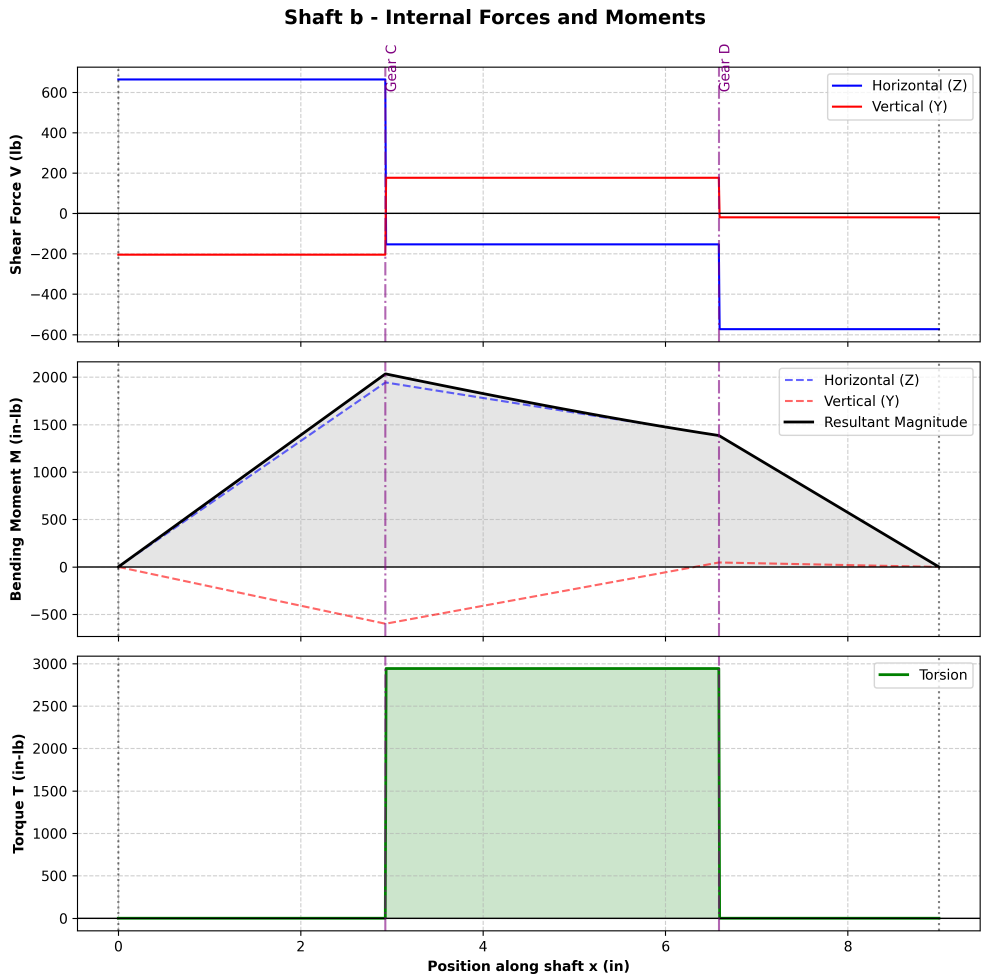

Figure 2: The shear, moment, and torsion diagrams for both the y and z planes for shaft b

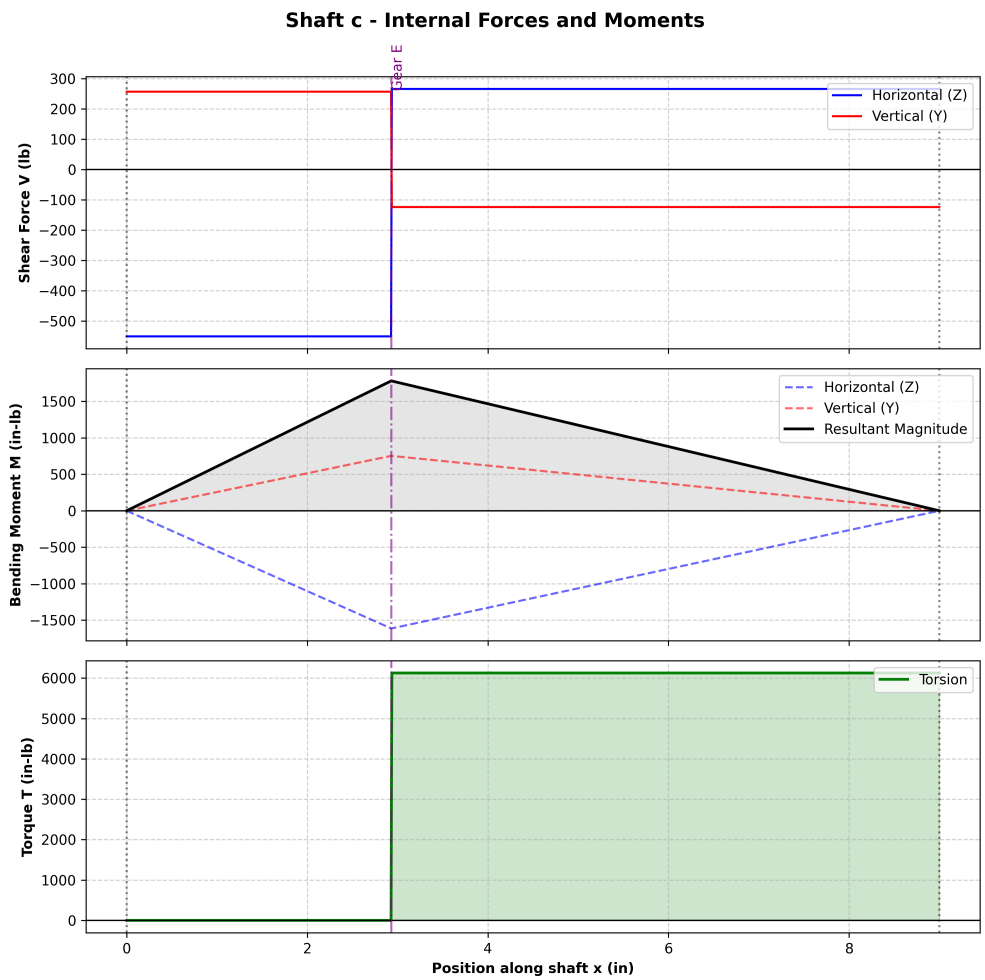

Figure 3: The shear, moment, and torsion diagrams for both the y and z planes for shaft c

These Functions and graphs can be used to evaluate the internal stresses of the shaft at the left and right side of each "extrema." As stated above, the transverse shear was ignored due to complexities of continuously or discreetly integrating over 90° to find a maximum every iteration. The internal stresses can then be turned into an equivalent stress using Von Mises (after splitting the stresses into the mean and alternating components).

$$\sigma_a = \frac{32M_{max}}{\pi d^3} \quad , \quad \sigma_m = \frac{16T_{max}}{\pi d^3}\sqrt{3} \tag{13}$$

The equivalent bending stress from Von Mises is found for both the mean and alternating by:

$$\sigma' = \sqrt{\sigma^2 + 3\tau^2} \tag{14}$$

3.2.3 Strength of Shaft

The Strength of the shaft for specifically bending (as the equivalent stress from Von Mises is a bending stress) is found by the following equation. Factors equal to one for this case are ignored.

$$S_n = (0.5S_u)C_L C_G C_S C_R \tag{15}$$

Once the Internal Stresses and Strengths are known they can be used to find the factor of safety for each extremum. For the sake of fair comparison, every extrema's stresses were considered and their factors were compared to find the global lowest factor. Only one of each locations factors was taken as the lowest as that ensured that no matter the type of failure, fatigue or yield, the weakest point was known. The lowest factor of safety is the "critical" location along the shaft.

$$\frac{1}{n_f} = \frac{\sigma_a}{S_n} + \frac{\sigma_m}{S_u} \quad , \quad n_y = \frac{S_y}{\sigma_a + \sigma_m} \tag{16}$$

3.2.4 Shaft Deflections and Separations

The shaft deflections (linear and angular) can be found using superposition of piecewise functions and evaluated at each item of interest. The example bellow follows the shaft b vertical derivation from previous steps.

$$\delta(x) = \begin{cases} \frac{Fbx}{6EIL}\left(x^2 + b^2 - L^2\right) & \text{for } 0 \leq x \leq a \\ \frac{Fa(L-x)}{6EIL}\left((L-x)^2 + a^2 - L^2\right) & \text{for } a \leq x \leq L \end{cases} \tag{17}$$

1. Deflection due to Gear C

Substituting the load, position $a_C = LL$, and distance to the right support $b_C = L - LL$:

$$\delta_{y,C}(x) = \begin{cases} \frac{-W_{rC} \cdot b_C \cdot x}{6EIL}\left(x^2 + b_C^2 - L^2\right) & \text{for } 0 \leq x \leq a_C \\ \frac{-W_{rC} \cdot a_C \cdot (L-x)}{6EIL}\left((L-x)^2 + a_C^2 - L^2\right) & \text{for } a_C \leq x \leq L \end{cases} \tag{18}$$

2. Deflection due to Gear D

Substituting the load, position $a_D = L - LR$, and distance to the right support $b_D = LR$:

$$\delta_{y,D}(x) = \begin{cases} \frac{W_{rD} \cdot b_D \cdot x}{6EIL}\left(x^2 + b_D^2 - L^2\right) & \text{for } 0 \leq x \leq a_D \\ \frac{W_{rD} \cdot a_D \cdot (L-x)}{6EIL}\left((L-x)^2 + a_D^2 - L^2\right) & \text{for } a_D \leq x \leq L \end{cases} \tag{19}$$

3. Total Vertical Superposition

The net vertical deflection along the continuous shaft is the sum of the individual components. Because the forces have opposite signs, their resulting deflections partially cancel each other out, demonstrating the stabilizing effect of alternating gear orientations:

$$\delta_y(x) = \delta_{y,C}(x) + \delta_{y,D}(x) \tag{20}$$

Once each component's location on a shaft was mapped to the deflection through the superposition function above. Total deflections (linear and angular) at gears and bearings can be found by the following vector equations below and evaluated against constraints.

$$\vec{\delta}_{mesh} = \vec{\delta}_{gear} + \vec{\delta}_{meshingGear} \quad , \quad \vec{\delta}_{mesh} = (\delta_{y,1} + \delta_{y,2})\hat{j} + (\delta_{z,1} + \delta_{z,2})\hat{k} \tag{21}$$

$$\Delta_{total} = \sqrt{(\Sigma\delta_y)^2 + (\Sigma\delta_z)^2} \quad , \quad \Theta_{mesh} = |\vec{\theta}_{gear}| = \sqrt{\theta_y^2 + \theta_z^2} \tag{22}$$

3.2.5 Critical Speed

The final constraint on the shaft is the critical speed. It is found by the simple equation bellow. It is important to note that only the deflections in the direction of the gravity vector are used (in my coordinate system y).

$$\omega_c = \sqrt{\frac{g\sum_{i=1}^{n}(w_i\delta_i)}{\sum_{i=1}^{n}(w_i\delta_i^2)}} \tag{23}$$

If the actual speed is half or less than the critical speed, then it passes.

3.3 Choosing Bearings from Catalogue

Given each shaft's diameter is known, a bearing can be selected which has a capacity exceeding the required life (same as the shaft it is mounted to, found from the input shaft life requirement gear train math), the force equivalent (which will just be the radial component as there is no axial force on any shaft), and the bore matching the shaft it is mounted on.

$$F_e = \sqrt{R_y^2 + R_z^2} \quad , \quad C_{req} = K_a \cdot F_e \left(\frac{L_{target}}{K_R \cdot L_R} \right)^{0.3} \tag{24}$$

If there is no bearing that meets the required capacity for that shaft's diameter, then the design is invalid.

4 Numerical Results & Validation Comparison

The Optimizer was paired with a validation script that took the project givens and the design parameters to independently calculate the results, comparing them on percent difference. This approach was used to guarantee no mistakes were made in the extensive program used to optimize the Gearbox.

4.1 Kinematics & Gear Mesh Forces

Pitch diameters are determined by the diametral pitch ($P_d = 10$ teeth/in) and number of teeth (N). Intermediate shaft (b) and output shaft (c) speeds and torques are derived from the input shaft (a) parameters ($\omega_{in} = 1050$ RPM, $T_{in} = 1260.5$ in-lb).

Explicit Numerical Calculation (Shaft b & Stage 1):

$$\begin{aligned}
d_B &= \frac{N_B}{P_d} = \frac{60}{10} = 6.0 \text{ in} \\
\omega_b &= \omega_{in} \left(\frac{N_B}{N_D} \right) = 1050 \left(\frac{60}{140} \right) = 450.00 \text{ RPM} \\
T_b &= T_{in} \left(\frac{N_D}{N_B} \right) = 1260.5 \left(\frac{140}{60} \right) = 2941.17 \text{ in-lb} \\
W_{t1} &= \frac{2T_{in}}{d_B} = \frac{2(1260.5)}{6.0} = 420.17 \text{ lb} \\
W_{r1} &= W_{t1} \tan(25^\circ) = 420.17 \tan(25^\circ) = 195.93 \text{ lb} \\
V_1 &= \frac{\pi d_B \omega_{in}}{12} = \frac{\pi(6.0)(1050)}{12} = 1649.34 \text{ ft/min}
\end{aligned}$$

4.2 AGMA Tooth Bending Stress

The fatigue strength (S_n) and AGMA bending stress (σ) determine the factor of safety (n).

Explicit Numerical Calculation (Gear B):

$$\begin{aligned}
S_n &= 0.5 S_u \cdot C_s \cdot K_r \cdot K_{ms} = 45.14 \text{ kpsi} \\
\sigma_B &= \frac{W_{t1} P_d}{F_B J_B} K_v K_o K_m = \frac{420.17(10)}{0.564 \cdot J_B} K_v K_o K_m = 45.14 \text{ kpsi} \\
n_B &= \frac{S_n}{\sigma_B} = \frac{45.14}{45.14} = 1.000
\end{aligned}$$

Parameter	Validated Calc	Optimizer Rep	Difference (%)
Shaft b Speed (RPM)	450.00	450.00	0.0
Shaft c Speed (RPM)	216.00	216.00	0.0
Shaft b Torque (in-lb)	2941.17	2941.20	0.0
Shaft c Torque (in-lb)	6127.43	6127.50	0.0
Stage 1 Tangential Force W_{t1} (lb)	420.17	420.20	0.0
Stage 1 Radial Force W_{r1} (lb)	195.93	195.90	0.0
Stage 1 Velocity V_1 (ft/min)	1649.34	1649.00	0.0
Stage 2 Tangential Force W_{t2} (lb)	816.99	817.00	0.0
Stage 2 Radial Force W_{r2} (lb)	380.97	381.00	0.0
Stage 2 Velocity V_2 (ft/min)	848.23	848.00	0.0

Component	Parameter	Validated Calc	Optimizer Rep	Difference (%)
Gear B	S_n (kpsi)	45.14	45.10	0.1
	σ (kpsi)	45.14	45.10	0.1
	FOS	1.000	1.000	0.0
Gear C	σ (kpsi)	45.12	45.10	0.0
	FOS	1.000	1.000	0.1
Gear D	σ (kpsi)	45.09	45.10	0.0
	FOS	1.001	1.000	0.1
Gear E	σ (kpsi)	45.12	45.10	0.0
	FOS	1.000	1.000	0.0

4.3 Exact Beam Statics & Deflections

Exact shaft deflections are validated using Macaulay's (singularity) functions to superimpose the discrete loading effects of each gear. The general deflection function $\delta(x)$ for a simply supported beam of length L with point loads P_i at locations a_i is evaluated independently for orthogonal planes:

$$\delta(x) = \frac{1}{EI}\left(\frac{R_L}{6}x^3 - \sum_i \frac{P_i}{6}\langle x - a_i\rangle^3 + C_1 x\right), \quad C_1 = \sum_i \frac{P_i(L-a_i)}{6L}\left((L-a_i)^2 - L^2\right)$$

Explicit Numerical Calculation (Shaft b Reaction & Deflection at Gear C, Horizontal z-plane):

$$EI = 29 \times 10^6 \left(\frac{\pi(1.575)^4}{64}\right) = 8.761 \times 10^6 \text{ lb-in}^2$$

$$R_{Lz} = \frac{816.99(9-2.93) + 420.17(9-6.587)}{9} = 663.67 \text{ lb}$$

$$C_{1z} = \frac{816.99(9-2.93)}{6(9)}\left[(9-2.93)^2 - 9^2\right] + \frac{420.17(9-6.587)}{6(9)}\left[(9-6.587)^2 - 9^2\right] = -6974.78$$

$$\delta_z(2.93) = \frac{1}{8.761 \times 10^6}\left(\frac{663.67}{6}(2.93)^3 - 0 - 0 - 6974.78(2.93)\right) = -0.0015 \text{ in} \Rightarrow |\delta_z| = 0.0015 \text{ in}$$

Location	Plane	Validated Calc (in)	Optimizer Rep (in)	Difference (%)
Shaft b, Gear C	Horizontal (z)	0.0015	0.0015	0.1
	Vertical (y)	0.0007	0.0007	6.4
Shaft b, Gear D	Horizontal (z)	0.0013	0.0013	0.2
	Vertical (y)	0.0006	0.0006	6.6
Shaft c, Gear E	Horizontal (z)	0.0011	0.0011	0.3
	Vertical (y)	0.0005	0.0004	11.7

4.4 Bearing Loads & Expected Life

Bearing reaction forces (F_e) drive the predicted cyclical life (L_{exp}).

Explicit Numerical Calculation (Bearing C, Left b):

$$F_e = \sqrt{R_{Lz}^2 + R_{Ly}^2} = \sqrt{663.67^2 + 309.47^2} = 732.27 \text{ lb}$$

$$L_{exp} = K_{r,brg}(90 \times 10^6)\left(\frac{C_{cat}}{K_a F_e}\right)^{10/3} = 1.0(90 \times 10^6)\left(\frac{2225}{1.0(732.27)}\right)^{10/3} = 223.12 \times 10^6 \text{ rev}$$

4.5 Shaft Fatigue & Yield Analysis

Maximum bending moments (M_{max}) govern the alternating stress (σ_a). The torque governs the mean stress (σ_m).

Bearing	Parameter	Validated Calc	Optimizer Rep	Difference (%)
C (Left b)	Total Reaction (lb)	732.27	727.70	0.6
	Angular Defl (deg)	0.0395	0.0391	0.9
	Expected Life (rev)	2.23×10^8	2.28×10^8	2.2
D (Right b)	Total Reaction (lb)	632.78	625.60	1.1
	Angular Defl (deg)	0.0371	0.0367	1.1
	Expected Life (rev)	3.63×10^8	3.78×10^8	4.1
E (Left c)	Total Reaction (lb)	607.98	597.10	1.8
	Angular Defl (deg)	0.0293	0.0287	2.0
	Expected Life (rev)	4.15×10^8	4.41×10^8	6.3
F (Right c)	Total Reaction (lb)	293.47	288.70	1.6
	Angular Defl (deg)	0.0232	0.0227	2.1
	Expected Life (rev)	4.66×10^8	4.93×10^8	5.9

Explicit Numerical Calculation (Shaft b):

$$M_{max} = \sqrt{(R_{Lz} \cdot x_C)^2 + (R_{Ly} \cdot x_C)^2} = \sqrt{(663.67 \cdot 2.93)^2 + (309.47 \cdot 2.93)^2} = 2145.3 \text{ in-lb}$$

$$\sigma_a = \frac{32M_{max}}{\pi d^3} = \frac{32(2145.3)}{\pi(1.575)^3} = 5.60 \text{ kpsi}$$

$$\sigma_m = \frac{16T_b\sqrt{3}}{\pi d^3} = \frac{16(2941.17)\sqrt{3}}{\pi(1.575)^3} = 6.64 \text{ kpsi}$$

$$n_{Goodman} = \left(\frac{\sigma_a}{S_{n,shaft}} + \frac{\sigma_m}{S_u}\right)^{-1} = \left(\frac{5.60}{21.93} + \frac{6.64}{74.0}\right)^{-1} = 3.92$$

$$n_{Yield} = \frac{S_y}{\sigma_a + \sigma_m} = \frac{65.4}{5.60 + 6.64} = 5.34$$

Shaft	Parameter	Validated Calc	Optimizer Rep	Difference (%)
Shaft b	Alt Stress (kpsi)	5.60	5.56	0.6
	Mean Stress (kpsi)	6.64	6.64	0.0
	FOS Goodman	3.92	3.93	0.4
	FOS Yield	5.34	5.35	0.2
Shaft c	Alt Stress (kpsi)	4.65	4.57	1.7
	Mean Stress (kpsi)	13.84	13.84	0.0
	FOS Goodman	3.39	3.42	0.9
	FOS Yield	3.53	3.55	0.4

4.6 System Constraints & Component Separation Check

Total separation distance limit ≤ 0.005 in. Angular misalignment $\leq 0.040°$. Component spacing limit ≥ 1.0 in. Component edge limits are defined by centroid location $\pm$ (width / 2).

Explicit Numerical Calculation (Separation & Interference):

$$\Delta_{Separation,Stage1} = \sqrt{(\delta_{z,GearD})^2 + (\delta_{y,GearD})^2} = \sqrt{(0.00126)^2 + (0.00055)^2} = 0.00138 \text{ in} \leq 0.005 \text{ in [PASS]}$$

$$\Delta_{BrgC,GearC} = \left(2.93 - \frac{0.952}{2}\right) - \left(0 + \frac{0.709}{2}\right) = 2.454 - 0.3545 = 2.100 \text{ in} \geq 1.0 \text{ in [PASS]}$$

$$\Delta_{GearC,GearD} = \left(6.587 - \frac{0.498}{2}\right) - \left(2.93 + \frac{0.952}{2}\right) = 6.338 - 3.406 = 2.932 \text{ in} \geq 1.0 \text{ in [PASS]}$$

$$\Delta_{GearD,BrgD} = \left(9.0 - \frac{0.709}{2}\right) - \left(6.587 + \frac{0.498}{2}\right) = 8.6455 - 6.836 = 1.810 \text{ in} \geq 1.0 \text{ in [PASS]}$$

$$\Delta_{BrgE,GearE} = \left(2.93 - \frac{0.851}{2}\right) - \left(0 + \frac{0.709}{2}\right) = 2.5045 - 0.3545 = 2.150 \text{ in} \geq 1.0 \text{ in [PASS]}$$

$$\Delta_{GearE,BrgF} = \left(9.0 - \frac{0.591}{2}\right) - \left(2.93 + \frac{0.851}{2}\right) = 8.7045 - 3.3555 = 5.349 \text{ in} \geq 1.0 \text{ in [PASS]}$$

4.7 Wholistic Analysis on Numerical Differences

It is evident that the program and validator script align until the deflections phase. I suspect the reason the vertical component of the deflection in particular suffered from larger percent differences was in rounding errors in the csv export which the validator script used to get its data. At the scale of hundred thousandths, these minor rounding errors show up as exaggerated larger percent differences. The horizontal components were made up of the larger tangential forces from the gears to they are an order of magnitude larger, exaggerating the different.

Another reason for this is the slight difference gravity makes in the calculations. The validator made standard "machine design course" assumptions of no gear mass effects, while the program took these into account. As the gravity makes up a minor part of the forces at play, the effect came out as 6-11% difference.

Moving on to the differences seen in bearing life, once again the mass of the gears create minor differences which are then exaggerated by the exponential scaling of the life equation calculation. Despite this, no life ever drops bellow the shaft's needed life to pass project specs.

5 Sources

- Surface Factor: Chap 08, Lecture 01, Slide 22
- Bearing Catalogue: Chap 14, Lecture 01, Slide 18
- Bearing Load Capacity Catalogue: Chap 14, Lecture 01, Slide 21
- Geometry Factor (gear): Chap 15, Lecture 02, Slide 11
- Reliability Factor (gear): Chap 15, Lecture 02, Slide 15
- Bearing Factors and Formulas: Chap 14, Lecture 01
- Shaft Deflections and Critical Speed: Chap 17
- Gear Formulas and Additional Factors: Chap 15 Lecture 02